Problem 1: Consider $S = \{\cos^2 x, \sin^2 x, \cos 2x\}$ that spans a vector space V.
(a) Show that S is not a basis. **(b)** Find a basis set containing $\cos 2x$.

Solution

(a) Since $\cos 2x = \cos^2 x - \sin^2 x$:

$$1 \cdot \cos^2 x + (-1) \cdot \sin^2 x + (-1) \cdot \cos 2x = 0.$$

The set is linearly dependent, hence not a basis.
(b) Given $V = \text{span}\{\cos^2 x, \sin^2 x, \cos 2x\}$, $\sin^2 x = \cos^2 x - \cos 2x$. By the plus/minus theorem, $S = \{\cos^2 x, \cos 2x\}$ has $\text{span}(S) = V$. For $a\cos^2 x + b\cos 2x = 0$, $\cos^2 x$ and $\cos 2x$ are not scalar multiples, so $a = b = 0$, making S a basis. Similarly, $\cos^2 x = \sin^2 x + \cos 2x$, so $T = \{\sin^2 x, \cos 2x\}$ has $\text{span}(T) = V$ and is linearly independent.

(a) Linearly dependent; (b) $\{\cos^2 x, \cos 2x\}$ or $\{\sin^2 x, \cos 2x\}$

Problem 2: Consider a set $S = \{\cos^2 x, \sin^2 x, \cos 2x\}$ that spans a vector space V.
(a) Show S is not a basis for V. **(b)** Find a basis for V containing $\sin^2 x$.

Solution

(a) Using $\cos 2x = \cos^2 x - \sin^2 x$:

$$1 \cdot \cos^2 x + (-1) \cdot \sin^2 x + (-1) \cdot \cos 2x = 0.$$

This non-trivial relation shows the set is linearly dependent, hence not a basis.
(b) Since $\cos 2x = \cos^2 x - \sin^2 x$, $\cos^2 x = \sin^2 x + \cos 2x$. Thus, $V = \text{span}\{\sin^2 x, \cos 2x\}$. For $a\sin^2 x + b\cos 2x = 0$, $x = 0$ gives $b = 0$; and $x = \pi/2$ gives $a = 0$. The set $\{\sin^2 x, \cos 2x\}$ is linearly independent and spans V, forming a basis.

(a) Linearly dependent; (b) $\{\sin^2 x, \cos 2x\}$

Problem 3: Consider the set $\left\{ \begin{pmatrix} 1 \\ 6 \\ 4 \end{pmatrix}, \begin{pmatrix} 2 \\ 4 \\ -1 \end{pmatrix}, \begin{pmatrix} -1 \\ 2 \\ 5 \end{pmatrix} \right\}$. Is it a basis for $\mathbb{R}^3$?

Solution

Form the matrix:

$$A = \begin{pmatrix} 1 & 2 & -1 \\ 6 & 4 & 2 \\ 4 & -1 & 5 \end{pmatrix}.$$

Compute the determinant:

$$\det(A) = 1 \begin{vmatrix} 4 & 2 \\ -1 & 5 \end{vmatrix} - 2 \begin{vmatrix} 6 & 2 \\ 4 & 5 \end{vmatrix} + (-1) \begin{vmatrix} 6 & 4 \\ 4 & -1 \end{vmatrix}$$

$$= 1(20 + 2) - 2(30 - 8) - 1(-6 - 16) = 22 - 44 + 22 = 0.$$

Since $\det(A) = 0$, the set is linearly dependent and not a basis.

$$\boxed{\text{No}}$$

Problem 4: Consider the set $\left\{ \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}, \begin{pmatrix} -2 \\ 1 \\ 4 \end{pmatrix}, \begin{pmatrix} -1 \\ 1 \\ 5 \end{pmatrix} \right\}$. Is it a basis for $\mathbb{R}^3$?

Solution

Form the matrix:

$$A = \begin{pmatrix} 1 & -2 & -1 \\ 0 & 1 & 1 \\ 1 & 4 & 5 \end{pmatrix}.$$

Compute the determinant:

$$\det(A) = 1 \begin{vmatrix} 1 & 1 \\ 4 & 5 \end{vmatrix} - (-2) \begin{vmatrix} 0 & 1 \\ 1 & 5 \end{vmatrix} + (-1) \begin{vmatrix} 0 & 1 \\ 1 & 4 \end{vmatrix}$$

$$= 1(5 - 4) + 2(0 - 1) - 1(0 + 1) = 1 - 2 + 1 = 0.$$

Since $\det(A) = 0$, the set is linearly dependent and not a basis.

$$\boxed{\text{No}}$$

Problem 5: Consider the set $\{1 - 3x + 2x^2, 1 + x + 4x^2, 1 - 7x\}$. Is it a basis for $\mathbb{P}_2$?

Solution

$k_1(1 - 3x + 2x^2) + k_2(1 + x + 4x^2) + k_3(1 - 7x) = 0$ gives the following system of linear equations in k_1, k_2, k_3.

$$\begin{aligned} k_1 + k_2 + k_3 &= 0 \\ -3k_1 + k_2 - 7k_3 &= 0 \\ 2k_1 + 4k_2 &= 0 \end{aligned}$$

Form the matrix:

$$A = \begin{pmatrix} 1 & 1 & 1 \\ -3 & 1 & -7 \\ 2 & 4 & 0 \end{pmatrix}.$$

Compute the determinant:

$$\det(A) = 1\begin{vmatrix} 1 & -7 \\ 4 & 0 \end{vmatrix} - 1\begin{vmatrix} -3 & -7 \\ 2 & 0 \end{vmatrix} + 1\begin{vmatrix} -3 & 1 \\ 2 & 4 \end{vmatrix}$$

$$= 1(0 + 28) - 1(0 + 14) + 1(-12 - 2) = 28 - 14 - 14 = 0.$$

Since $\det(A) = 0$, the system have non-trivial solution. Hence the set is linearly dependent and not a basis.

$$\boxed{\text{No}}$$

Problem 6: Consider the set $\{4+6x+x^2, -1+4x+2x^2, 5+2x-x^2\}$. Is it a basis for $\mathbb{P}_2$?

Solution

$k_1(4+6x+x^2)+k_2(-1+4x+2x^2)+k_3(5+2x-x^2)=0$ gives the following system of linear equations in k_1, k_2, k_3.

$$\begin{aligned} 4k_1 - k_2 + 5k_3 &= 0 \\ 6k_1 + 4k_2 + 2k_3 &= 0 \\ k_1 + 2k_2 - k_3 &= 0 \end{aligned}$$

Form the matrix:

$$A = \begin{pmatrix} 4 & -1 & 5 \\ 6 & 4 & 2 \\ 1 & 2 & -1 \end{pmatrix}.$$

Compute the determinant:

$$\det(A) = 4\begin{vmatrix} 4 & 2 \\ 2 & -1 \end{vmatrix} - (-1)\begin{vmatrix} 6 & 2 \\ 1 & -1 \end{vmatrix} + 5\begin{vmatrix} 6 & 4 \\ 1 & 2 \end{vmatrix}$$

$$= 4(-8) + 1(-8) + 5(8) = -32 - 8 + 40 = 0.$$

Since $\det(A) = 0$, the set is linearly dependent and not a basis.

No

Problem 7: Consider the set $\left\{ \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}, \begin{pmatrix} -2 \\ 1 \\ 4 \end{pmatrix}, \begin{pmatrix} 0 \\ 3 \\ 1 \end{pmatrix} \right\}$. Is it a basis for $\mathbb{R}^3$?

Solution

Form the matrix:

$$A = \begin{pmatrix} 1 & -2 & 0 \\ 0 & 1 & 3 \\ 1 & 4 & 1 \end{pmatrix}.$$

Compute the determinant:

$$\det(A) = 1\begin{vmatrix} 1 & 3 \\ 4 & 1 \end{vmatrix} - (-2)\begin{vmatrix} 0 & 3 \\ 1 & 1 \end{vmatrix} + 0\begin{vmatrix} 0 & 1 \\ 1 & 4 \end{vmatrix}$$

$$= 1(1-12) + 2(0-3) = -11 - 6 = -17.$$

Since $\det(A) \neq 0$, the set is linearly independent and a basis.

Yes

Problem 8: Consider the set $\left\{ \begin{pmatrix} 2 \\ 3 \\ 1 \end{pmatrix}, \begin{pmatrix} 4 \\ 1 \\ 1 \end{pmatrix}, \begin{pmatrix} 0 \\ -7 \\ 1 \end{pmatrix} \right\}$. Is it a basis for $\mathbb{R}^3$?

Solution

Form the matrix:

$$A = \begin{pmatrix} 2 & 4 & 0 \\ 3 & 1 & -7 \\ 1 & 1 & 1 \end{pmatrix}.$$

Compute the determinant:

$$\det(A) = 2 \begin{vmatrix} 1 & -7 \\ 1 & 1 \end{vmatrix} - 4 \begin{vmatrix} 3 & -7 \\ 1 & 1 \end{vmatrix} + 0 \begin{vmatrix} 3 & 1 \\ 1 & 1 \end{vmatrix}$$

$$= 2(1+7) - 4(3+7) = 16 - 40 = -24.$$

Since $\det(A) \neq 0$, the set is linearly independent and a basis.

$$\boxed{\text{Yes}}$$